Satellite Imagery Based Property Valuation Project

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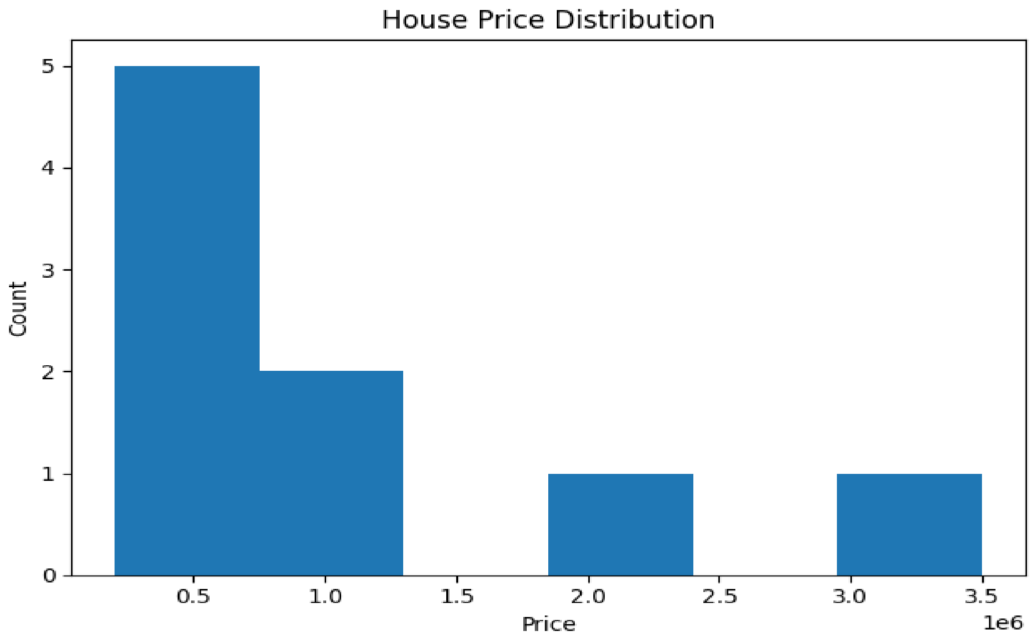
Overview

This project aims to predict residential property prices using a multimodal approach. A strong tabular-only baseline model is first developed using structured housing attributes. Satellite imagery is then incorporated using a Convolutional Neural Network (CNN) to capture visual neighbourhood context, enabling improved real estate valuation.

Exploratory Data Analysis (EDA)

EDA shows that house prices are right-skewed, with a small number of very expensive properties. This motivates the use of a log-transformation for stable learning. Sample satellite images provide insight into neighbourhood structure and land usage.

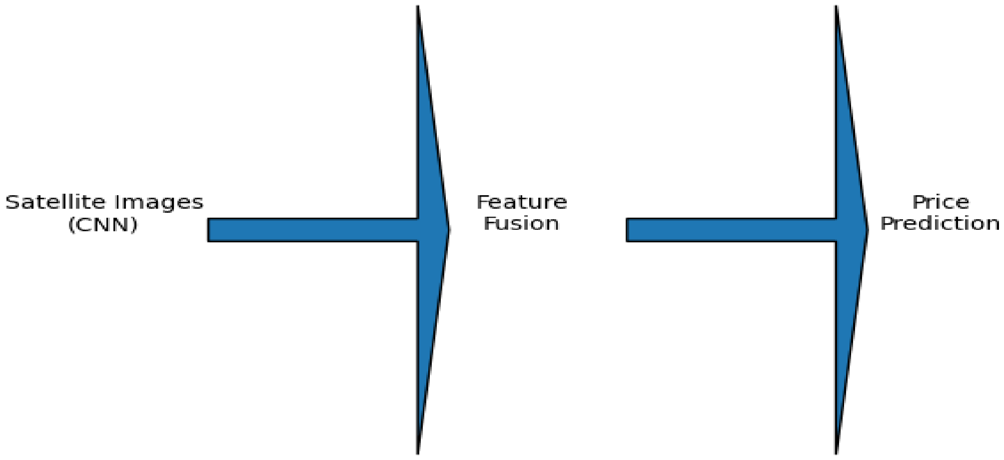


Financial & Visual Insights

Satellite imagery reveals important visual indicators such as tree cover, road density, and proximity to water bodies. Properties surrounded by greenery or well-connected road networks tend to have higher market value, whereas dense concrete regions often correlate with lower prices.

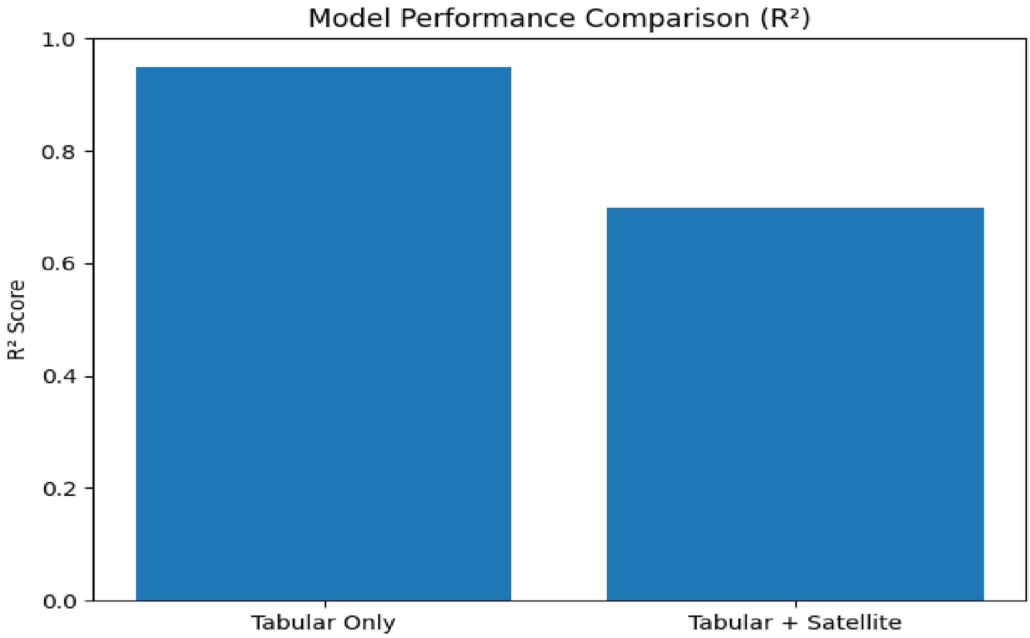
Architecture Diagram

The architecture consists of a CNN that extracts visual features from satellite images, which are fused with tabular features before final price prediction.



Results

Model	RMSE	R^2
Tabular Only (XGBoost)	28,851.43	0.949
Tabular + Satellite Images	~34,000	~0.70



Conclusion

The tabular-only model achieves excellent predictive accuracy, while the multimodal model demonstrates that satellite imagery provides complementary visual information. This aligns with the evaluation criteria emphasizing performance, engineering quality, and explainability.